In this framework, it is indeed possible to compare spaces through *isotopy equivalence*, a stronger notion than homeomorphism: $X \subseteq \mathbb{R}^d$ and $Y \subseteq \mathbb{R}^d$ are said to be *(ambient) isotopic* if there exists a continuous family of homeomorphisms $H : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$, H continuous, such that for any $t \in [0,1]$, $H_t = H(t,.) : \mathbb{R}^d \to \mathbb{R}^d$ is an homeomorphism, H_0 is the identity map in $\mathbb{R}^d$ and $H_1(X) = Y$. Obviously, if X and Y are isotopic, then they are homeomorphic. The converse is not true: a knotted and an unknotted circles in $\mathbb{R}^3$ are not homeomorphic (notice that although this claim seems rather intuitive, its formal proof requires the use of some non obvious algebraic topology tools).

4.1 Distance-like functions and reconstruction

Given a compact subset K of $\mathbb{R}^d$, and a non negative real number r, the union of balls of radius r centered on K, $K^r = \cup_{x \in K} B(x,r)$, called the r-offset of K, is the r-sublevel set of the distance function $d_K : \mathbb{R}^d \to \mathbb{R}$ defined by $d_K(x) = \inf_{y \in K} \|x - y\|$; in other words, $K^r = d_k^{-1}([0,r])$. This remark allows to use differential properties of distance functions and to compare the topology of the offsets of compact sets that are close to each other with respect to the Hausdorff distance.

Definition 3 (Hausdorff distance in $\mathbb{R}^d$)**.** *The Hausdorff distance between two compact subsets K, K' of $\mathbb{R}^d$ is defined by*

$$d_H(K, K') = \|d_K - d_{K'}\|_\infty = \sup_{x \in \mathbb{R}^d} |d_K(x) - d_{K'}(x)|.$$

In our setting, the considered compact sets are the data set $\mathbb{X}_n$ and of the support M of the measure μ. When M is a smooth compact submanifold, under mild conditions on $d_H(\mathbb{X}_n, M)$, for some well-chosen r, the offsets of $\mathbb{X}_n$ are homotopy equivalent to M, Chazal and Lieutier (2008a); Niyogi et al. (2008) - see Figure 5 for an illustration. These results extend to larger classes of compact sets and leads to stronger results on the inference of the isotopy type of the offsets of M, Chazal et al. (2009c,d). They also lead to results on the estimation of other geometric and differential quantities such as normals Chazal et al. (2009c), curvatures Chazal et al. (2008) or boundary measures Chazal et al. (2010) under assumptions on the Haussdorff distance between the underlying shape and the data sample.

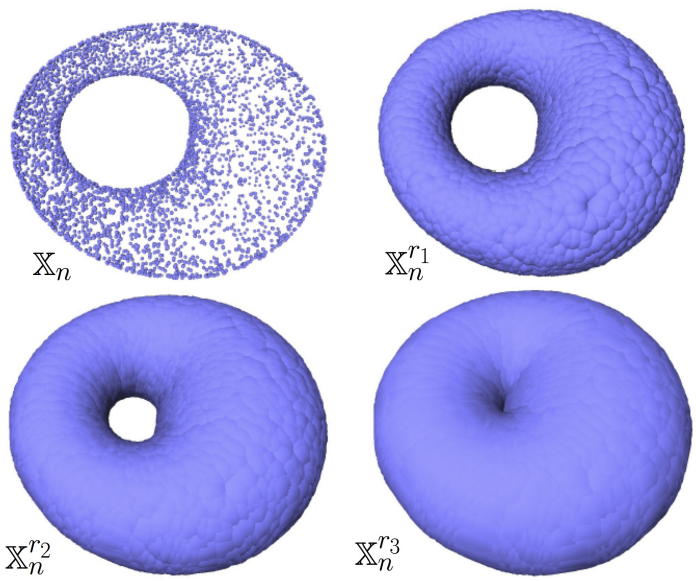

Figure 5: The example of a point cloud $\mathbb{X}_n$ sampled on the surface of a torus in $\mathbb{R}^3$ (top left) and its offsets for different values of radii $r_1 < r_2 < r_3$. For well chosen values of the radius (e.g. r_1 and r_2), the offsets are clearly homotopy equivalent to a torus.

These results rely on the 1-semiconcavity of the squared distance function d_K^2, i.e. the convexity of the function $x \to \|x\|^2 - d_K^2(x)$, and can be naturally stated in the following general framework.

Definition 4. *A function $\phi : \mathbb{R}^d \to \mathbb{R}_+$ is* distance-like *if it is proper (the pre-image of any compact set in $\mathbb{R}$ is a compact set in $\mathbb{R}^d$) and $x \to \|x\|^2 - \phi^2(x)$ is convex.*

Thanks to its semiconcavity, a distance-like function ϕ have a well-defined, but not continuous, gradient $\nabla\phi : \mathbb{R}^d \to \mathbb{R}^d$ that can be integrated into a continuous flow (Petrunin, 2007) that allows to track the evolution of the topology of its sublevel sets and to compare it to the one of the sublevel sets of close distance-like functions.

Definition 5. *Let ϕ be a distance-like function and let $\phi^r = \phi^{-1}([0, r])$ be the r-sublevel set of ϕ.*

- *A point $x \in \mathbb{R}^d$ is called* α-critical *if $\|\nabla_x \phi\| \leq \alpha$. The corresponding value $r = \phi(x)$ is also said to be* α-critical.
- *The* weak feature size *of ϕ at r is the minimum $r' > 0$ such that ϕ does not have any critical value between r and $r + r'$. We denote it by $\mathrm{wfs}_\phi(r)$. For any $0 < \alpha < 1$, the* α-reach *of ϕ is the maximum r such that $\phi^{-1}((0, r])$ does not contain any α-critical point.*

The weak feature size $\mathrm{wfs}_\phi(r)$ (resp. α-reach) measures the regularity of ϕ around its r-level sets (resp. O-level set). When $\phi = d_K$ is the distance function to a compact set $K \subset \mathbb{R}^d$, the 1-reach coincides with the classical reach from geometric measure theory Federer (1959). Its estimation from random samples has been studied in Aamari et al. (2019). An important property of a distance-like function ϕ is the topology of their sublevel sets ϕ^r can only change when r crosses a 0-critical value.

Lemma 1 (Isotopy Lemma Grove (1993))**.** *Let ϕ be a distance-like function and $r_1 < r_2$ be two positive numbers such that ϕ has no 0-critical point, i.e. points x such that $\nabla\phi(x) = 0$, in the subset $\phi^{-1}([r_1, r_2])$. Then all the sublevel sets $\phi^{-1}([0, r])$ are isotopic for $r \in [r_1, r_2]$.*

As an immediate consequence of the Isotopy Lemma, all the sublevel sets of ϕ between r and $r + \mathrm{wfs}_\phi(r)$ have the same topology. Now the following reconstruction theorem from Chazal et al. (2011b) provides a connection between the topology of the sublevel sets of close distance-like functions.

Theorem 2 (Reconstruction Theorem)**.** *Let ϕ, ψ be two distance-like functions such that $\|\phi - \psi\|_\infty < \varepsilon$, with $\mathrm{reach}_\alpha(\phi) \geq R$ for some positive ε and α. Then, for every $r \in [4\varepsilon/\alpha^2, R - 3\varepsilon]$ and every $\eta \in (0, R)$, the sublevel sets ψ^r and ϕ^η are homotopy equivalent when*

$$\varepsilon \leq \frac{R}{5 + 4/\alpha^2}.$$

Under similar but slightly more technical conditions the Reconstruction Theorem can be extended to prove that the sublevel sets are indeed homeomorphic and even isotopic Chazal et al. (2009c, 2008).

Coming back to our setting, and taking for $\phi = d_M$ and $\psi = d_{\mathbb{X}_n}$ the distance functions to the support M of the measure μ and to the data set $\mathbb{X}_n$, the condition $\mathrm{reach}_\alpha(d_M) \geq R$ can be interpreted as regularity condition on M[6]. The Reconstruction Theorem combined with the Nerve Theorem tell that, for well-chosen values of r, η, the η-offsets of M are homotopy equivalent to the nerve of the union of balls of radius r centered on $\mathbb{X}_n$, i.e the Cech complex $\mathrm{Cech}_r(\mathbb{X}_n)$.

From a statistical perspective, the main advantage of these results involving Hausdorff distance is that the estimation of the considered topological quantities boil down to support estimation questions that have been widely studied - see Section 4.3.

[6]As an example, if M is a smooth compact submanifold then $\mathrm{reach}_0(\phi)$ is always positive and known as the reach of M Federer (1959).

4.2 Homology inference

The above results provide a mathematically well-founded framework to infer the topology of shapes from a simplicial complex built on top of an approximating finite sample. However, from a more practical perspective it raises raise two issues. First, the Reconstruction Theorem requires a regularity assumption through the α-reach condition that may not always be satisfied and, the choice of a radius r for the ball used to build the Čech complex $\mathrm{Cech}_r(\mathbb{X}_n)$. Second, $\mathrm{Cech}_r(\mathbb{X}_n)$ provides a topologically faithfull summary of the data, through a simplicial complex that is usually not well-suited for further data processing. One often needs easier to handle topological descriptors, in particular numerical ones, that can be easily computed from the complex. This second issue is addressed by considering the homology of the considered simplicial complexes in the next paragraph, while the first issue will be addressed in the next section with the introduction of persistent homology.

Homology in a nutshell. Homology is a classical concept in algebraic topology providing a powerful tool to formalize and handle the notion of topological features of a topological space or of a simplicial complex in an algebraic way. For any dimension k, the k-dimensional "holes" are represented by a vector space H_k whose dimension is intuitively the number of such independent features. For example the 0-dimensional homology group H_0 represents the connected components of the complex, the 1-dimensional homology group H_1 represents the 1-dimensional loops, the 2-dimensional homology group H_2 represents the 2-dimensional cavities,...

To avoid technical subtleties and difficulties, we restrict the introduction of homology to the minimum that is necessary to understand its usage in the following of the paper. In particular we restrict to homology with coefficients in $\mathbb{Z}_2$, i.e. the field with two elements 0 and 1 such that $1 + 1 = 0$, that turns out to be geometrically a little bit more intuitive. However, all the notions and results presented in the sequel naturally extend to homology with coefficient in any field. We refer the reader to Hatcher (2001) for a complete and comprehensible introduction to homology and to Ghrist (2017) for a recent concise and very good introduction to applied algebraic topology and its connections to data analysis.

Let K be a (finite) simplicial complex and let k be a non negative integer. The *space of k-chains* on K, $C_k(K)$ is the set whose elements are the formal (finite) sums of k-simplices of K. More precisely, if $\{\sigma_1, \cdots \sigma_p\}$ is the set of k-simplices of K, then any k-chain can be written as

$$c = \sum_{i=1}^{p} \varepsilon_i \sigma_i \quad \text{with} \quad \varepsilon_i \in \mathbb{Z}_2.$$

If $c' = \sum_{i=1}^{p} \varepsilon'_i \sigma_i$ is another k-chain and $\lambda \in \mathbb{Z}_2$, the sum $c + c'$ is defined as $c + c' = \sum_{i=1}^{p} (\varepsilon_i + \varepsilon'_i)\sigma_i$ and the product $\lambda.c$ is defined as $\lambda.c = \sum_{i=1}^{p} (\lambda.\varepsilon_i)\sigma_i$, making $C_k(K)$ a vector space with coefficients in $\mathbb{Z}_2$. Since we are considering coefficient in $\mathbb{Z}_2$, geometrically a k-chain can be seen as a finite collection of k-simplices and the sum of two k-chains as the symmetric difference of the two corresponding collections[7].

The *boundary* of a k-simplex $\sigma = [v_0, \cdots, v_k]$ is the $(k-1)$-chain

$$\partial_k(\sigma) = \sum_{i=0}^{k} (-1)^i [v_0, \cdots, \hat{v}_i, \cdots, v_k]$$

where $[v_0, \cdots, \hat{v}_i, \cdots, v_k]$ is the $(k-1)$-simplex spanned by all the vertices except v_i [8]. As the k-simplices form a basis of $C_k(K)$, ∂_k extends as a linear map from $C_k(K)$ to $C_{k-1}(K)$ called the *boundary operator*. The kernel $Z_k(K) = \{c \in C_k(K) : \partial_k = 0\}$ of ∂_k is called the *space of k-cycles of K* and the image $B_k(K) = \{c \in C_k(K) : \exists c' \in C_{k+1}(K), \partial_{k+1}(c') = c\}$ of ∂_{k+1} is

[7] Recall that the symmetric difference of two sets A and B is the set $A \Delta B = (A \setminus B) \cup (B \setminus A)$.

[8] Notice that as we are considering coefficients in $\mathbb{Z}_2$, here $-1 = 1$ and thus $(-1)^i = 1$ for any i.